Final Presentation Plan: Does Air Quality Explain Adult Asthma?

Overview

- **Audience:** TruBridge Health Analytics reviewers and healthcare stakeholders who want to know whether air quality should guide asthma prevention. Mixed technical background.
- **Goal:** Communicate one clear finding and what to do next, in 7 slides.
- **Core message:** County-level air quality does not predict differences in adult asthma across the 769 counties analyzed. State matters far more, and county averages hide neighborhood differences, so decisions need neighborhood-level data.
- **Approach:** Insight-first. The answer appears on slide 1. There is no slide per analysis: the statistical tests are merged into one finding.

Three ideas the presentation keeps separate

1. **What the analysis found:** no statistically significant association between county-level AQI and adult asthma prevalence.
2. **What else it showed:** state and county size explain far more of the variation than AQI does.
3. **What it suggests:** healthcare organizations should combine environmental, demographic, and neighborhood-level data instead of relying on county AQI alone.

Storyline

1. County-level air quality does not predict differences in adult asthma prevalence
2. We expected cleaner counties to report less adult asthma, so we tested it across 769 U.S. counties
3. Counties with worse air do not report more asthma, and no test finds a significant difference
4. State matters far more than air quality in explaining county asthma
5. Los Angeles has above-median AQI and below-average asthma, which shows the limits of county averages
6. These results show the limits of county data, not that air pollution is harmless
7. Combining air quality with neighborhood and demographic data is the next step for targeting asthma prevention

Slide Details

Slide 1: Title and answer

- **Action title:** County-level air quality does not predict differences in adult asthma prevalence

- **Body:**
  - Subtitle: a county-level analysis of CDC and EPA data
  - Two large numbers: 10.67% adult asthma in the lowest-AQI quartile vs. 10.61% in the highest-AQI quartile
  - Presenter, externship, and date
- **Visual:** two large number callouts side by side, one for the lowest-AQI quartile and one for the highest-AQI quartile
- **Speaker notes:** Lead with the answer. The gap is 0.06 percentage points. This describes the 769 counties analyzed, not the effect of air pollution on any one person.

Slide 2: Expectation, question, and data

- **Action title:** We expected cleaner counties to report less adult asthma, so we tested it across 769 U.S. counties
- **Body:**
  - Expectation: air pollution is a recognized respiratory health concern, and the Los Angeles area has long faced air-quality challenges
  - Data: CDC PLACES (modeled adult asthma, from an earlier survey year) joined to EPA median AQI (2026, partial year)
  - Reading AQI: a higher AQI means poorer air quality
  - Method: correlation, group comparisons, and regression with state effects
- **Visual:** four-box flow: CDC PLACES and EPA AirData, then merge, then 769 counties, then analysis
- **Speaker notes:** Keep this to about one minute. The two datasets cover different time periods. Sample details are in backup slide B3.

Slide 3: Main finding

- **Action title:** Counties with worse air do not report more asthma, and no test finds a significant difference
- **Body:**
  - Correlation between median AQI and asthma prevalence: r = -0.04
  - Lowest vs. highest AQI quartile: 10.67% vs. 10.61%
  - Additional tests and regression also find no significant association (details in backup)
- **Visual:** Figure 2, a scatter plot with labeled axes and a flat trend line, with Los Angeles County highlighted in orange
- **Speaker notes:** The point is the flat line. These results show association, not causation. The tests are not independent confirmations, because they examine the same data in different ways. The full test list is in backup slide B2.

Slide 4: State and county size

- **Action title:** State matters far more than air quality in explaining county asthma
- **Body:**
  - AQI alone explains 0.3% of the variation in county asthma
  - Adding county size raises that to 9%, and adding state raises it to 67% (about 65% adjusted for the number of variables)

    - The highest county averages are in Maine, Oregon, West Virginia, Massachusetts, and Oklahoma
- **Visual:** simple three-bar chart of explained variation: AQI alone 0.3%, plus county size 9%, plus state 67%. Label it as model fit, not causal effects.
- **Speaker notes:** This is association, not causation. State differences may reflect demographics, socioeconomic conditions, healthcare access, geography, and how PLACES models small areas (its model includes state and county effects). These are possible explanations, not findings. Larger counties also report slightly less asthma (r = -0.26), shown in backup slide B1.

Slide 5: The Los Angeles area

- **Action title:** Los Angeles has above-median AQI and below-average asthma, which shows the limits of county averages
- **Body:**
    - Los Angeles County: AQI 62 (higher than 99% of the 769 counties, but "Moderate" on the EPA scale) and asthma 9.1% vs. a 10.6% average
    - Orange (9.0%) and San Diego (8.9%) look similar, while Trinity and Humboldt have the highest asthma in California (11.0%)
    - Los Angeles research shows asthma clusters by neighborhood, which a county average hides
- **Visual:** Figure 5, California counties with Los Angeles highlighted
- **Speaker notes:** Los Angeles illustrates the limits of county averages. It does not show that AQI is unrelated to asthma within the county. The median AQI across the 769 counties is 41.

Slide 6: Limitations

- **Action title:** These results show the limits of county data, not that air pollution is harmless
- **Body:**
    - Geographic: county averages hide neighborhood exposure
    - Missing variables: income, race and ethnicity, and smoking are not in the data
    - Measurement: AQI is not long-term exposure, and PLACES covers adults only
    - Sample: only monitored counties are included (827 records, 769 analyzed), which skews toward larger counties
- **Visual:** three simple columns (geography, missing variables, measurement) with a short sample note underneath
- **Speaker notes:** Say plainly that this is a limit of the data, not evidence of safety. Association is not causation: larger counties reporting less asthma does not mean population growth reduces asthma.

Slide 7: Recommendations and next steps

- **Action title:** Combining air quality with neighborhood and demographic data is the next step for targeting asthma prevention
- **Body:**
    - County averages hide local differences: analyze asthma and pollution at the census-tract level in Los Angeles County

    - State differences explain a lot: investigate demographic, socioeconomic, and geographic differences across states
    - Income and demographics are missing: add income, poverty, and race and ethnicity from the American Community Survey
    - Adult asthma is the only outcome: run a separate childhood asthma analysis with multi-year PM2.5 data
- **Visual:** two-column table pairing each finding with its next step
- **Takeaway callout:** Healthcare organizations should combine environmental, demographic, and neighborhood-level data, and should not target asthma programs by county AQI alone.
- **Speaker notes:** End on the takeaway, then invite questions and suggestions for neighborhood-level air quality sources.

Backup Slides (Not Counted)

- **B1, Distributions and county size:** Figure 1 (asthma averages 10.6%, median AQI is 41) and Figure 4 (asthma vs. county population on a log scale, r = -0.26).
- **B2, Test details:** Figure 3 (box plots by AQI quartile) and Table 1 (summary of all tests and p-values, from 0.19 to 0.82).
- **B3, Sample and data cleaning:** 827 AQI records + 3 extra rows from records that matched both a city and a county - 27 unmatched records - 34 matched counties with no asthma estimate = 769 rows. Removing the three double matches does not change the results.
- **B4, Largest counties:** In the top quarter of counties by population (n = 193), the correlation is r = 0.05 (p = 0.54) and AQI is not significant in regression (p = 0.075). Among the 42 counties with 1 million or more residents, r = 0.19 (p = 0.24). Neither is statistically significant, but both lean positive, which supports a finer-grained follow-up.
- **B5, Model details:** The 9% model predicts age-adjusted asthma from median AQI and the log of county population. The 67% model adds state effects (45 states) with standard errors clustered by state. R-squared values are in-sample, and the adjusted value for the state model is about 65%.

Design Notes

- One action title and one visual per slide, with at most four supporting points.
- Reuse the report's figures with a consistent palette: blue for data, orange for Los Angeles.
- Slide 4 needs one new chart with three values: 0.3%, 9%, and 67%.
- Put "Source: CDC PLACES, EPA AirData (2026), author analysis" in the footer of each data slide.
- Slides 1 and 3 carry the answer, so no separate executive summary slide is needed.